VAPT PRACTICAL REPORT

Cyber Security Internship – Week 5

Nmap • Nikto • Burp Suite

Submitted by	Amar Pratap
Organization	DG Interns Hub
Target	testphp.vulnweb.com
Environment	Kali Linux / Burp Suite
Scope	Authorized internship lab

Scope note: The screenshots in this report are the real practical screenshots supplied for this Week 5 submission. Testing is presented as authorized lab activity only.

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Week 5 – Vulnerability Assessment & Penetration Testing (VAPT)

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Evidence included: Nmap scans, Nikto troubleshooting, Burp Suite interception, request inspection and HTTP History using the real screenshots supplied for the practical.

Submitted by: Amar Pratap
Organization: DG Interns Hub

Target: testphp.vulnweb.com

1. Objectives & Workflow

• Use Nmap to discover hosts, ports and services.

• Attempt web-server checks with Nikto.

• Intercept and inspect HTTP requests using Burp Suite.

• Document actual output and limitations without inventing findings.

Practical workflow

Reconnaissance → Service identification → Web assessment attempt → HTTP interception → Evidence-based interpretation.

2. Nmap – Version & SYN Scan

Nmap 7.99 was verified in Kali Linux. The SYN scan showed the authorized target as reachable and identified 53/tcp as open.

```
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.4 libssh2-1.11.1 libz-1.3.1 libpcre2-10.48 libpcap-1.10.7 nmap-libdnet-1.18.
0 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

Figure 1. Real Nmap version output.

```
└─$ nmap -sS testphp.vulnweb.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-06 22:14 +0530
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.010s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
53/tcp open  domain

Nmap done: 1 IP address (1 host up) scanned in 5.36 seconds
```

Figure 2. Real SYN scan output.

3. Nmap – Service & Aggressive Scan

Service detection identified 53/tcp as dnsmasq 2.80. The aggressive scan could not determine a reliable OS match because the required open/closed TCP-port combination was not available.

```
└─$ nmap -sV testphp.vulnweb.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-06 22:20 +0530
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.0012s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
53/tcp open  domain  dnsmasq 2.80

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 58.08 seconds
```

Figure 3. Real Nmap service/version detection.

```
└─$ nmap -A testphp.vulnweb.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-06 22:22 +0530
Nmap scan report for testphp.vulnweb.com (44.228.249.3)
Host is up (0.0014s latency).
rDNS record for 44.228.249.3: ec2-44-228-249-3.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE    VERSION
53/tcp open  tcpwrapped
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host

TRACEROUTE (using port 80/tcp)
HOP RTT    ADDRESS
1   ... 30

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 49.10 seconds
```

Figure 4. Real Nmap aggressive scan.

4. Nikto – Web Assessment Attempt

Nikto 2.6.1 was installed successfully. The scan could not connect to the target on port 80.



Figure 5. Real Nikto version output.

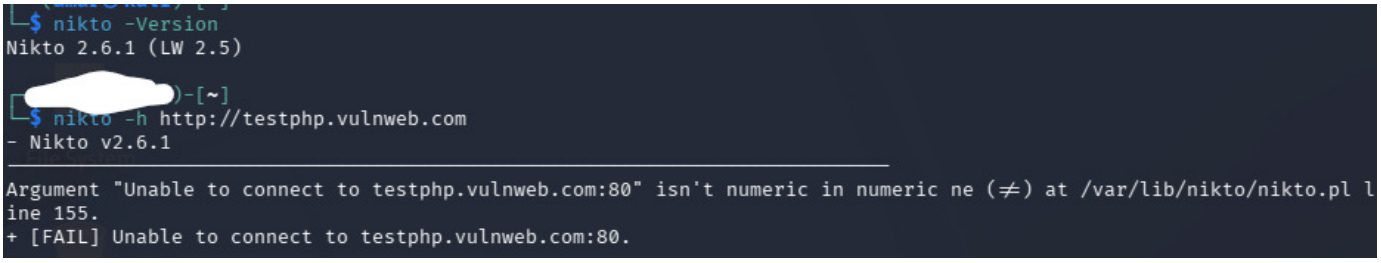


Figure 6. Real Nikto failure with Nmap/curl connectivity evidence.

> Interpretation: Nmap reported 80/tcp and 443/tcp as filtered, while curl also failed to connect to port 80. This is recorded as a connectivity/availability limitation, not as a confirmed vulnerability.

5. Burp Suite – HTTP Interception

Burp Suite successfully captured HTTP requests from the authorized lab target. The real screenshots show Intercept enabled and the request details visible in the lower panel.

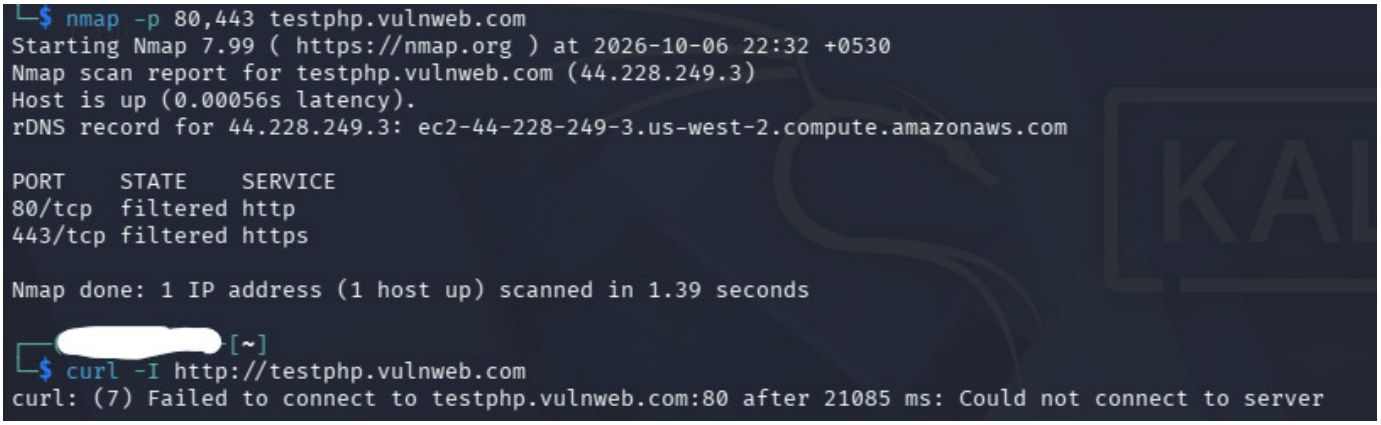


Figure 7. Real Burp interception screenshot.

6. Burp Suite – Request Inspection

The captured artists.php request shows the HTTP method, Host header, User-Agent, Accept headers and query parameters.

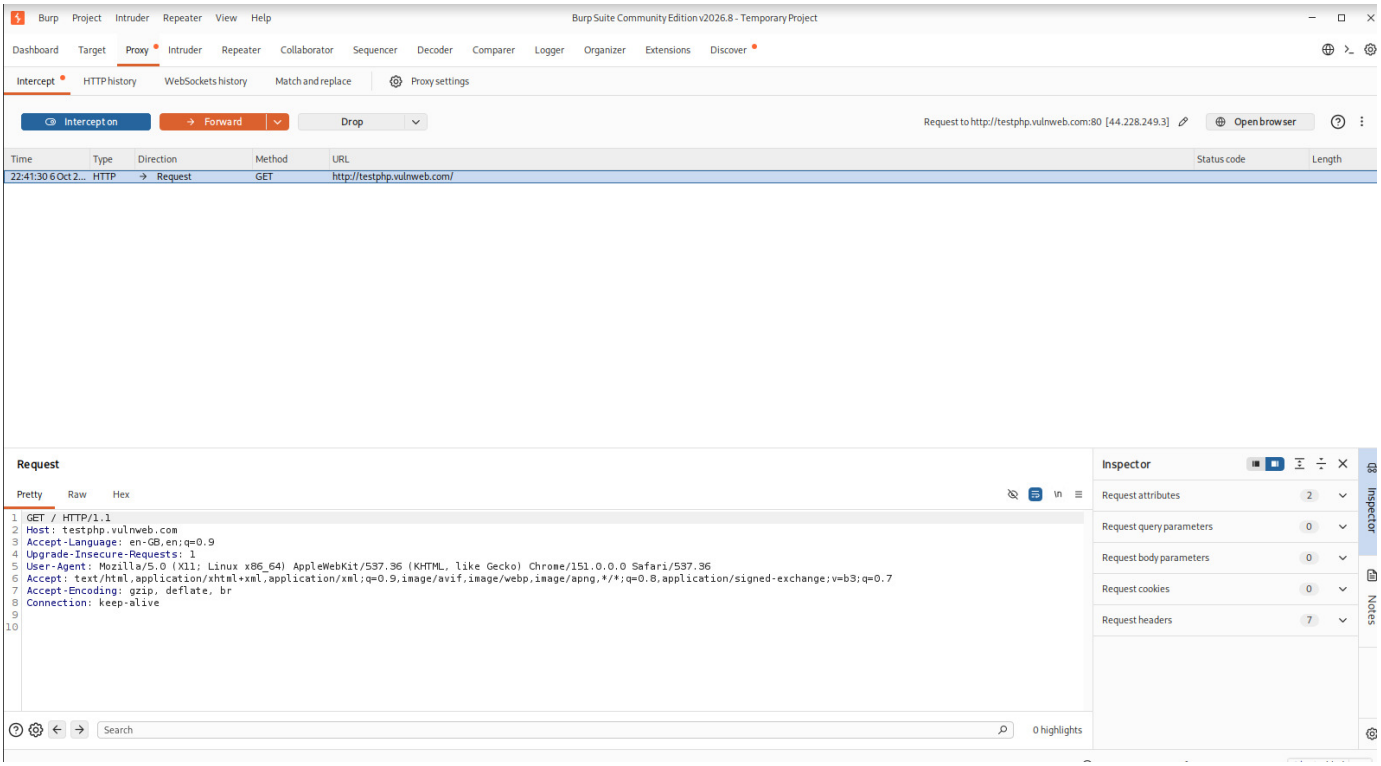


Figure 8. Real artists.php request.

7. Burp Suite – Additional Request

The categories.php request was also captured and inspected. This demonstrates successful interception of multiple application endpoints.

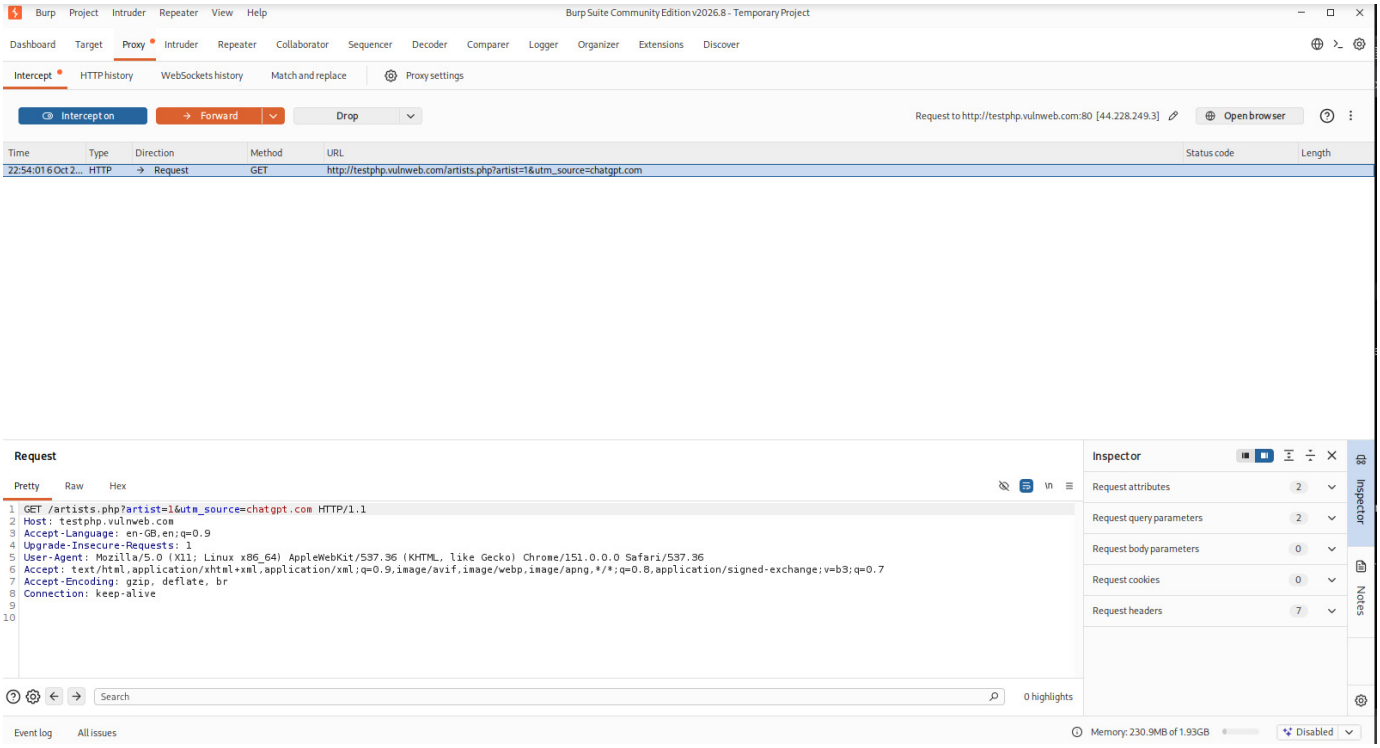


Figure 9. Real categories.php request.

8. Burp Suite – HTTP History

HTTP History provided a consolidated record of repeated GET requests and the captured PHP endpoints.

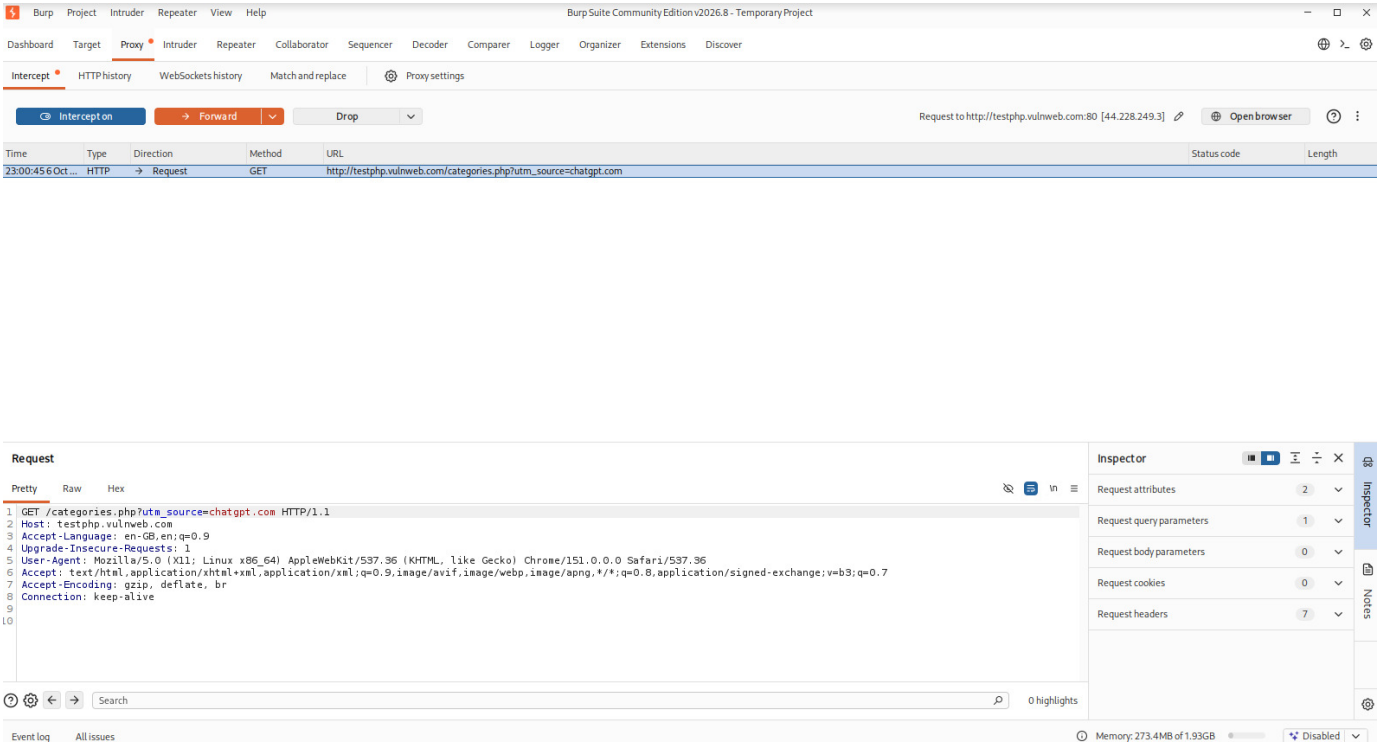


Figure 10. Real Burp HTTP History.

9. Findings & Interpretation

Area	Evidence	Interpretation
Nmap	53/tcp open; dnsmasq 2.80	Observed exposed service; not automatically a vulnerability.
Nikto/HTTP	80/443 filtered; curl failed	Web assessment was limited by connectivity.
Burp Suite	GET requests and parameters visible	Successful HTTP interception and inspection.

No unsupported SQL injection, XSS, authentication bypass, RCE or other high-impact vulnerability is claimed. The report stays limited to what the supplied evidence demonstrates.

10. Challenges, Learning & Conclusion

• Nikto could not reach the web service; Nmap and curl were used to verify the condition.

• Burp interception and HTTP History confirmed that application requests were being captured.

• The main learning was to validate scanner output and distinguish an observation from a confirmed vulnerability.

• Completed practical areas: Nmap scans, Nikto setup/troubleshooting, Burp interception, request inspection and HTTP History.

Conclusion

Week 5 provided practical exposure to reconnaissance, service enumeration, web assessment troubleshooting and HTTP traffic analysis. The report uses the real screenshots supplied from the lab session and records the limitations honestly.

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